

184. Department Highest Salary

PLATFORM

Platform: LeetCode

DIFFICULTY

MEDIUM

DATE

Solved: June 10, 2026

Problem Summary

Write a solution to find employees who have the highest salary in each of the departments.

Return the result table in **any order**.

The result format is in the following example.

Approach (Subquery + Join)

- Find the maximum salary for each department.
- Join employees with departments.
- Keep only employees whose salary equals their department's maximum salary.
- Return department name, employee name, and salary.

SQL Concepts Used

- JOIN
- GROUP BY
- MAX()
- Derived Table (Subquery)

Key Insights

- Highest salary must be found **within each department**, not globally.
- We need to compare an employee's salary with the maximum salary of their department.
- Multiple employees may have the same highest salary.
- Department name comes from the `Department` table, so a JOIN is required.

Algorithm

- Group employees by department.
- Compute maximum salary for each department.
- Join this result back with Employee.
- Match employees whose salary equals the department maximum.
- Join Department table to get department names.
- Return required columns.

Time Complexity & Space Complexity

TC : $O(n)$

SC : $O(d)$

Database

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Medium

Topics

Companies

SQL Schema

Pandas Schema

Table: Employee

Column Name	Type
id	int
name	varchar
salary	int
departmentId	int

id is the primary key (column with unique values) for this table.

departmentId is a foreign key (reference columns) of the ID from the Department table.

Each row of this table indicates the ID, name, and salary of an employee. It also contains the ID of their department.

Table: Department

Column Name	Type
id	int
name	varchar

id is the primary key (column with unique values) for this table. It is guaranteed that department name is not NULL.

Each row of this table indicates the ID of a department and its name.

Write a solution to find employees who have the highest salary in each of

2.4K

146

54 Online

Accepted

All Submissions

Accepted 14 / 14 testcases passed

Sahil Khan submitted at Jun 10, 2026 14:17

Runtime

617 ms | Beats 95.64%

343ms729ms907ms1084ms

Code | MySQL

```
1 SELECT d.name AS Department, e.name AS Employee, e.salary AS Salary
2 FROM Employee e
3 INNER JOIN Department d
4 ON e.departmentId = d.id
5 WHERE (e.salary, e.departmentId) IN (
6     SELECT MAX(salary), departmentId nvj
7     FROM Employee
8     GROUP BY departmentId
9 )
```

Write your notes here

Select related tags

0/5

Code

MySQL

```
1 SELECT d.name AS Department, e.name AS Employee, e.salary AS Salary
2 FROM Employee e
3 INNER JOIN Department d
4 ON e.departmentId = d.id
5 WHERE (e.salary, e.departmentId) IN (
6     SELECT MAX(salary), departmentId nvj
7     FROM Employee
8     GROUP BY departmentId
9 )
```

Saved Upgrade to Cloud Saving

Ln 1, Col 1

Testcase

Test Result

You must run your code first